

Yuelin Liu

Full-Stack Developer | AI/LLM Product Engineering

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Full Australian work rights - Temporary Graduate visa (subclass 485), valid until March 2029

Summary

Early-career full-stack developer building AI-assisted product/workflow systems, developer tooling and reliable backend/data applications across TypeScript, Node.js, Next.js and Python. Projects combine messy real-world inputs with structured local state, bounded model orchestration, deterministic validation/domain logic, configurable human/provider controls and testable delivery paths. Early-stage SaaS experience includes OpenAI function calling, BFF/REST integration across web and mobile clients, Postman/Jest testing, NGINX/CI/CD exposure, instrumentation and backend performance profiling.

Experience

Independent Developer

Sep 2025 – Present

Prospective Client Demos for a Melbourne Consultancy

Melbourne, Australia

- Translated informal briefs for prospective specialty-coffee and drone-training clients into working authenticated, database-backed demonstrations.
- Built the prototypes with Next.js 15 App Router, Clerk OAuth and Convex, adapting scope and implementation as requirements clarified.

Full-Stack Developer

Mar 2025 – Jun 2025

ByteCroniX - Early-Stage AI SaaS

Melbourne, Australia

- Implemented OpenAI function-calling flows in a Node.js/Express codebase for AI-assisted product interactions, working across BFF routes and REST integrations supporting web, Android and iOS clients.
- Worked across API testing and delivery workflows using Postman, Jest/Supertest, NGINX configuration/troubleshooting and multi-service CI/CD, contributing alongside the wider development team.
- Added timing instrumentation to an inherited points-summary path and identified scoring services as the dominant latency contributors.
- Profiled an inherited points-summary pipeline and parallelised independent service calls, reducing an observed local integration-test response from 26.7s to 5.6s.

Selected Product & Engineering Systems

Job Search Dispatch | Local AI-assisted job-search operations system

JavaScript, Node.js, browser extension, LLM APIs

- Built a local workflow that turns live job pages and candidate evidence into persistent application tickets, combining job ingestion, structured candidate/evidence state, AI-assisted analysis and drafting, application lifecycle tracking and human approval.
- Structured the system around a browser/public-ATS ingestion layer, file-backed Canonical Assets with confirmation and usage scopes, separate Triage/Pack AI processing, and evidence-constrained drafting rather than letting conversation history become the source of truth.
- Added BYOK multi-provider control (Anthropic, DeepSeek, OpenAI, Gemini or custom), per-workflow model selection, trace diagnostics, server-owned timeouts and upstream cancellation while preserving the accepted zero-dependency, zero-build-step local app baseline.

MoneyGuard | Local-first wage and cashflow decision support for hourly workers

TypeScript, Node.js, Next.js 16, React 19, Gemini, DeepSeek, Zod, Dexie

- Built a system connecting timecard input to weekly wage/burn/surplus insight, labour-value simulation, buffer and shift-impact planning, payday forecasting, delayed-payday comparison and local audit history.
- Separated responsibilities across extraction, validation, domain logic and AI: Gemini structured OCR → Zod validation → deterministic wage/finance calculations → optional explanation/reasoning, with provider interfaces and deterministic mock/fixture paths.
- Kept structured financial history behind Dexie/IndexedDB storage adapters and isolated optional hosted OCR/explanation behind narrower server contracts, bounded inputs and separately configured provider paths.

AI Harness | GitHub-native control plane for AI-assisted software delivery

GitHub, Skills, shell, deterministic verification

- Built a model-neutral delivery workflow designed to reduce manual coordination and context drift by moving one approved product outcome from durable GitHub work state through implementation, exact-candidate review and repository release.
- Separated shared lifecycle contracts/governance from stage-specific Skills; binds human acceptance to the exact pull-request head SHA and moves repeatable facts into scripts, tests and CI instead of conversation memory or parallel status systems.

DSH Conversation Exporter v0.3 | AI conversation portability and context handoff

JavaScript, Node.js, DSH plugin APIs, Markdown

- Built a local DSH Web plugin that turns the current session into clean, Git-friendly Markdown for reading and cross-assistant handoff, supporting full export or selected whole turns without replacing DSH's raw Session Log.
- Performs host-side semantic filtering before export, preserving Markdown/Unicode while excluding reasoning, tool activity and runtime metadata; package verification covers tests, syntax checks and npm dry-run. Listed in the Awesome DSH Plugin catalogue.

Additional Engineering

Melbourne University Ultimate Club Platform - Built a Next.js 15/TypeScript/MongoDB single-club operations platform spanning public, member and admin workflows for roster/tournament selection, events, announcements, alumni and video; later centralised server-side authorisation around database-backed roles and turned write-path defects into regression tests plus GitHub Actions typecheck/lint/test/build checks.

Client Data Delivery Validator - Built a bounded Python/pandas CSV data-quality pipeline with six validation rule families, structured `ValidationIssue/ValidationResult` output and 12 pytest regression cases.

Technical Skills

Languages & Querying: TypeScript, JavaScript, Python, SQL

AI & Validation: OpenAI function calling, Gemini, DeepSeek, Amazon Bedrock, Zod, structured model outputs, deterministic validation boundaries

Frontend: React 19, Next.js 15/16, Tailwind CSS

Backend & Data: Node.js, Express, REST APIs, BFF patterns, MongoDB, Convex, pandas, IndexedDB/Dexie

Testing & API Tools: Postman, Jest, Supertest, Vitest, Playwright, pytest, React Testing Library

Cloud & Delivery: Git/GitHub workflows, GitHub Actions, Docker, NGINX, CI/CD; exposure to Kubernetes/GKE; AWS Lambda, SQS, Bedrock, Aurora Serverless v2, Terraform

Education & Applied AI Architecture

The University of Melbourne

2022 – 2023

Master of Information Technology

Melbourne, Australia

University of Nottingham Ningbo China

2016 – 2020

BSc (Hons) Computer Science

Ningbo, China

Agentic AI Engineering Course Implementation on AWS (Ed Donner, Udemy), completed Nov 2025.

Adapted the ALEX capstone into an SQS-backed multi-role workflow with Terraform-defined AWS architecture using Bedrock, Lambda, SQS and Aurora Serverless v2; deployed and later decommissioned the study environment when its running cost outweighed its learning value.

Technical Writing

Weighted Nearest Neighbours via Metric Learning and Batch Gradient Descent - sole-authored undergraduate dissertation with analytical MSE gradient derivation and a reusable Python framework evaluated on three UCI datasets.

Lattice-Based Post-Quantum Cryptography: Key Agreement Protocols on the LWE Problem - three-author University of Melbourne course paper; authored the lattice preliminaries and LWE problem formulations.